

Database Design

Updated 2026-07-13 17:56. This PDF documents the current schema in database/schema.sql. No database additions or migrations were applied while updating this file.

Executive Summary

The schema uses PostgreSQL with pgvector and pgcrypto. It supports authentication, refresh-token sessions, conversations, messages, uploaded document metadata, document ownership/scoping, document text chunks, and vector embeddings.

Important: this document lists the actual tables and attributes currently defined in database/schema.sql. A separate drift section notes fields the backend code expects but that are not currently defined in the SQL schema.

Extensions And Enum

Object	Definition	Purpose
Extension	vector	Enables VECTOR(1536) storage and vector similarity search.
Extension	pgcrypto	Provides gen_random_uuid() for UUID defaults.
Enum	message_role: system, user, assistant	Restrains message roles to known chat roles.

Table Overview

Table	Purpose
users	Stores account identity and login credentials.
user_sessions	Stores refresh-token sessions for login validity, expiry, rotation, and logout.
conversations	Stores chat threads owned by users.
messages	Stores messages inside conversations.
documents	Stores uploaded document metadata.
user_documents	Join table linking users to documents they own or can access.
conversation_documents	Join table scoping documents to a conversation for retrieval.
document_chunks	Stores readable text chunks extracted from documents.
chunk_embeddings	Stores pgvector embeddings for document chunks.

Physical Tables And Attributes

users

Stores account identity and login credentials.

Attribute	Type	Details
id	UUID	Primary key. Defaults to gen_random_uuid().
username	VARCHAR(50)	Required. Unique username.
email	VARCHAR(255)	Required. Unique email.
password_hash	TEXT	Required. Stores hashed password, not raw password.
created_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- idx_users_email on email

user_sessions

Stores refresh-token sessions for login validity, expiry, rotation, and logout.

Attribute	Type	Details
id	UUID	Primary key. Defaults to gen_random_uuid().
user_id	UUID	Required. References users(id). Deletes cascade.
refresh_token	TEXT	Required. Unique token stored server-side.
device	VARCHAR(255)	Optional device/browser label.
ip_address	VARCHAR(100)	Optional IP address string.
is_revoked	BOOLEAN	Required. Defaults to false.
expires_at	TIMESTAMPTZ	Required. Session is invalid after this time.
created_at	TIMESTAMPTZ	Required. Defaults to now().
updated_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- idx_user_sessions_refresh on refresh_token

conversations

Stores chat threads owned by users.

Attribute	Type	Details
id	UUID	Primary key. Defaults to gen_random_uuid().
user_id	UUID	Required. References users(id). Deletes cascade.
title	VARCHAR(255)	Required. Defaults to 'New conversation'.
created_at	TIMESTAMPTZ	Required. Defaults to now().
updated_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- idx_conversations_user on user_id

messages

Stores messages inside conversations.

Attribute	Type	Details
id	UUID	Primary key. Defaults to gen_random_uuid().
conversation_id	UUID	Required. References conversations(id). Deletes cascade.
role	message_role	Required enum: system, user, or assistant.
content	TEXT	Required message text.
created_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- idx_messages_conversation on conversation_id, created_at

documents

Stores uploaded document metadata.

Attribute	Type	Details
id	UUID	Primary key. Defaults to gen_random_uuid().
name	VARCHAR(255)	Required original/display name.
storage_path	TEXT	Required server path for the stored file.
mime_type	VARCHAR(100)	Required file MIME type.
file_size	BIGINT	Required file size in bytes.
created_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- None defined in schema.sql.

user_documents

Join table linking users to documents they own or can access.

Attribute	Type	Details
user_id	UUID	Required. References users(id). Deletes cascade.
document_id	UUID	Required. References documents(id). Deletes cascade.
created_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- Primary key: user_id, document_id
- idx_user_documents_doc on document_id

conversation_documents

Join table scoping documents to a conversation for retrieval.

Attribute	Type	Details
conversation_id	UUID	Required. References conversations(id). Deletes cascade.
document_id	UUID	Required. References documents(id). Deletes cascade.
added_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- Primary key: conversation_id, document_id
- idx_convdocs_document on document_id

document_chunks

Stores readable text chunks extracted from documents.

Attribute	Type	Details
id	UUID	Primary key. Defaults to gen_random_uuid().
document_id	UUID	Required. References documents(id). Deletes cascade.
chunk_number	INTEGER	Required sequence number inside the document.
chunk_text	TEXT	Required extracted text for this chunk.
created_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- Unique constraint: document_id, chunk_number
- idx_chunks_document on document_id

chunk_embeddings

Stores pgvector embeddings for document chunks.

Attribute	Type	Details
id	UUID	Primary key. Defaults to gen_random_uuid().
chunk_id	UUID	Required. Unique. References document_chunks(id). Deletes cascade.
embedding	VECTOR(1536)	Required 1536-dimensional semantic vector.
created_at	TIMESTAMPTZ	Required. Defaults to now().

Indexes / constraints:

- idx_chunk_embeddings_hnsw using hnsw on embedding vector_cosine_ops

Relationships

From	To	Behavior
user_sessions.user_id	users.id	ON DELETE CASCADE
conversations.user_id	users.id	ON DELETE CASCADE
messages.conversation_id	conversations.id	ON DELETE CASCADE
user_documents.user_id	users.id	ON DELETE CASCADE
user_documents.document_id	documents.id	ON DELETE CASCADE
conversation_documents.conversation_id	conversations.id	ON DELETE CASCADE
conversation_documents.document_id	documents.id	ON DELETE CASCADE
document_chunks.document_id	documents.id	ON DELETE CASCADE
chunk_embeddings.chunk_id	document_chunks.id	ON DELETE CASCADE

How The RAG Tables Work

documents stores file metadata. document_chunks stores readable pieces of each document. chunk_embeddings stores a 1536-number vector for each chunk. During chat, the app embeds the user's question and compares it with chunk_embeddings to find the most relevant document_chunks.

Schema Drift Observed

The backend currently references additional fields/tables that are not present in database/schema.sql. These are documented here only; this PDF update does not add them to the database.

Expected by code	Where it appears	Why it matters
users.profile_picture, users.updated_at	auth/users routers and UserRead	Returned in auth and user API responses.
conversations.model_name, system_prompt, generation_config, is_archived	conversation/message routers	Conversation model settings and archive state.
messages.parent_message_id, model_name, token_count, generation_time, is_helpful, feedback_text	messages router and schemas	Branching, response metadata, and feedback.
messages.search_vector	conversation search endpoint	Full-text conversation search.
documents.description, content_hash, status, uploaded_at	documents router and DocumentRead	Deduplication and ingestion status tracking.
document_chunks.page_number, token_count	document processing and retrieval	Source location and chunk metadata.
embedding_models, chunk_embeddings.embedding_model_id	document/message routers	Tracks which embedding model produced vectors.
message_citations	messages router	Stores retrieved sources for assistant answers.
password_reset_tokens	auth router	Password reset flow.